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$$\int \cos(\ln x)dx$$

$$\begin{cases} u = \cos(\ln x) \\ dv = dx \end{cases} \Rightarrow \begin{cases} du = \dfrac{-\sin(\ln x)}{x}dx \\ v = x \end{cases}$$

$$\Rightarrow x\cos(\ln x) + \int x \cdot \frac{\sin(\ln x)}{x}dx$$

$$\begin{cases} u = \sin(\ln x) \\ dv = dx \end{cases} \Rightarrow \begin{cases} du = \dfrac{\cos(\ln x)}{x} \\ v = x \end{cases}$$

$$\Rightarrow x\cos(\ln x) + x\sin(\ln x) - \int x \cdot \frac{\cos(\ln x)}{x}dx$$

Breng $\int \cos(\ln x)dx$ naar het linkerlid

$$2\int \cos(\ln x)dx = x\cos(\ln x) + x\sin(\ln x) + c$$

$$\Rightarrow \int \cos(\ln x)dx = \frac{x}{2}\cos(\ln x) + \frac{x}{2}\sin(\ln x) + c$$

References